

The Mechanical Vacuum: Frame-Dragging, Satiation Velocity, and the Variable Gravitational Constant

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Abstract

In standard General Relativity (GR), G is a coupling constant that determines how much spacetime curves for a given amount of energy-momentum. GR treats G as a static scalar, whereas this model treats it as a dynamic equilibrium state of the field itself. We suggest an **Equilibrium Perspective**: the constant G is the restorative response of the vacuum field. As the matter-energy density ($T_{\mu\nu}$) increases, the vacuum impedance rises, effectively damping the field's ability to respond. This paper derives G as a function of vacuum restorative stiffness and matter-energy density, providing a mechanical solution for flat galactic rotation curves and the observed Newtonian limit in high-density solar systems.

1 The Mechanical Basis of Gravitation

In standard relativistic astrophysics, G is treated as a universal constant. However, applying a mechanical framework to the vacuum suggests that G represents the coupling efficiency between matter and the elastic fabric of space.

We define c^2 as the restorative stiffness of the vacuum medium. Matter-energy density ρ_m acts as the mechanical load. This paper builds upon the derivation for the gravitational constant G as a function of vacuum restorative stiffness and density²:

$$G = \frac{c^2}{8\pi\rho_m} \quad (1)$$

where ρ_m is interpreted as an effective linear mass density (mass per unit length, M/L). This allows the units to align ($m^3 \cdot kg^{-1} \cdot s^{-2}$) and provides a geometric reason for suppression: as matter “packs” into a region, the field's ability to be further displaced (its “tensor strength”) reaches a saturation point.

2 Tensor Saturation and the Newtonian Limit

In high-density systems like the Solar System, the presence of a massive central body increases the local tensor stiffness. To prevent G from hitting zero (which would halt gravitational interaction), we define the effective gravitational coupling as an asymptotic approach to the Newtonian floor:

$$G(\rho_m) = G_{vac} \cdot \exp\left(-\frac{\rho_{local}}{\rho_{crit}}\right) + G_{Newton} \quad (2)$$

where ρ_{crit} is a “Critical Saturation Density.” On galactic scales, the vacuum remains “pliant,” allowing for a higher effective G . This enhancement provides the additional centripetal force required to maintain flat rotation curves at large radii without the need for cold dark matter.

3 Derivation of Velocity-Gravity Equivalence

The core of this framework lies in the link between the gravitational coupling constant (G) and the propagation velocity of light (v_γ).

3.1 Defining Vacuum Refractive Index

Applying a mechanical load ρ_m to the vacuum medium, we can model the refractive index n_v based on the density of the tensor field:

$$n_v = \sqrt{1 + \rho_m} \quad (3)$$

Substituting our expression for ρ_m from Equation (1):

$$\rho_m = \frac{c^2}{8\pi G} \quad (4)$$

3.2 Deriving the Phase Velocity

Substituting Equation (4) into the velocity equation $v_\gamma = c/n_v$:

$$v_\gamma = \frac{c}{\sqrt{1 + \frac{c^2}{8\pi G}}} = c \sqrt{\frac{8\pi G}{8\pi G + c^2}} \quad (5)$$

3.3 Numerical Evaluation

Using standard SI values ($c \approx 2.9979 \times 10^8$ m/s, $G \approx 6.6743 \times 10^{-11}$ m³ kg⁻¹ s⁻²), the ratio within the square root is approximately 1.866×10^{-26} . Taking the root and multiplying by c :

$$v_\gamma \approx 0.00004 \text{ m/s} \quad (6)$$

This value identifies the propagation floor or “viscosity” of the local vacuum frame.

4 The Mechanical Integration of Screening Mechanisms

The observed suppression of G in high-density environments is not unique to this model; it aligns with established screening theories in modified gravity. By viewing the vacuum as a medium with variable **tensor susceptibility**, we can integrate these mechanisms under a single mechanical framework.

4.1 Chameleon Screening: Density-Driven Displacement

In the Chameleon mechanism, the scalar field’s mass is a function of the ambient density. In our framework, this is interpreted as a non-linear change in the vacuum’s **effective elastic modulus**. In low-density regions, the vacuum exhibits high **tensor susceptibility**, allowing matter to displace the field and create a long-range gradient (the fifth force). In high-density regions (Solar System), the field is **saturated** by the high ambient matter-energy density. This increases the **tensor stiffness**, rendering the vacuum unresponsive to additional displacement. The force is suppressed because the field has reached its maximum displacement capacity.

4.2 Symmetron Theory: Symmetry as Structural Phase

The Symmetron model relies on a density-dependent phase transition. We interpret this as a transition in the coupling efficiency of the vacuum field. At high densities, the field resides in a symmetric state where its interaction cross-section with matter is effectively zero; consequently, the tensor field remains unperturbed, and no additional displacement occurs. In low-density environments, spontaneous symmetry breaking occurs as the field shifts to a non-zero vacuum expectation value. The vacuum enters a high-susceptibility phase, where the matter-field coupling is restored. This mechanical displacement of the field manifests as the observed "fifth force" or anomalous acceleration in galactic outskirts.

4.3 The DGP Model: Dimensional Leakage and Bottlenecking

The Dvali-Gabadadze-Porrati (DGP) model traditionally suggests that gravitational flux "leaks" into a higher-dimensional bulk at infra-red scales. In this mechanical framework, we reinterpret this leakage as the vacuum's recovery of its ground-state tensor strength in low-density environments. Under the **Vainshtein Screening** effect, near massive sources, the tensor field reaches its **satiated limit**, where G is clamped to the 4D Newtonian value. This satiation effectively "shuts the door" on the extra-dimensional scaling, forcing the vacuum to behave as a rigid 4D manifold. Consequently, the transition scale r_c in DGP is recontextualized as the density threshold where the vacuum shifts from a satiated, high-impedance state (Solar System) to an unsatiated, high-susceptibility state (Deep Space).

5 Validation and Scaling: The Triple-Satiation Proof

We define Mechanical Satiation as the state wherein the vacuum's tensor susceptibility reaches a non-linear limit due to high ambient matter-energy density. In this satiated regime, the gravitational constant G is suppressed to its minimum Newtonian value, effectively 'clamping' the field and preventing the manifestation of modified gravity effects observed in lower-density, unsatiated environments. The transition from local suppression to universal maxima is governed by the dynamic range of the vacuum impedance.

We define the Satiation Ratio $\mathcal{S} = v_\gamma/c$, which represents the scaling factor between the local propagation floor and the maximum vacuum wave speed. The propagation floor, v_γ , is the minimum phase velocity of a wave in a mechanically satiated field—specifically, the velocity at which the vacuum impedance, driven by local matter-energy density ρ_m , reaches its maximum effective stiffness. By correlating c to the maximum propagation speed of a wave in a zero-density field (G_{max}), we define v_γ as the characteristic velocity of the local frame where the gravitational coupling is clamped to the Newtonian floor (G_{Newton})."

In the context of the $G = \frac{c^2}{8\pi\rho_m}$ theory, the Propagation Floor refers to the minimum permissible phase velocity of a wave within the vacuum medium when it is under maximum mechanical load. To understand this physically, we must view the vacuum not as an empty void, but as a mechanical field with a finite capacity for displacement.

5.1 The Variable Wave Speed Concept: Definition of Satiation

In classical wave mechanics, the speed of a wave (v) is determined by the ratio of the medium's stiffness (C) to its density (ρ): $v = \sqrt{C/\rho}$. In this model, c represents the Maximum Propagation Speed—the velocity of a wave in a ground-state vacuum where density ρ_m is negligible and the field is at its most elastic. The Propagation Floor (v_γ) is the velocity of that same wave when the density ρ_m has increased to the point of Mechanical Satiation. At this limit, the vacuum impedance is so high that the field becomes "stiff," slowing the propagation of the gravitational/electromagnetic wave to its absolute lowest possible value.

5.2 The "Clamping" Mechanism

The floor acts as a non-linear bottleneck. As we add matter to the system (increasing ρ_m), the wave speed drops. However, the vacuum cannot become infinitely stiff; it hits a physical limit where the tensor field can no longer be compressed. This limit is the Propagation Floor. In the Solar System, we are living in a region where the vacuum is "clamped" at this floor. This is why G appears to be a constant here—we are at the bottom of the curve where the field is satiated and no longer fluctuates.

5.3 Relation to the Satiation Ratio ($\mathcal{S}$)

The ratio $\mathcal{S} = v_\gamma/c$ defines the dynamic range of the vacuum. If $\mathcal{S} \approx 1$, the vacuum is unsatiated (Deep Space), and G is at its maximum. If $\mathcal{S} \approx 0.00004$ (as defined here), the vacuum is satiated (Solar System), and G is at its Newtonian floor.

5.4 Empirical Context

The reason we call it a "floor" is that it represents the ground-state jitter or "viscosity" of the local frame. By identifying $v_\gamma \approx 40 \mu\text{m/s}$, we suggest that this is the slowest "speed of gravity/light" possible in our local high-density environment. This speed correlates to the detected "frame-dragging" values by experiments like Gravity Probe B.

5.5 5.2 Local Evidence: Mechanical Satiation (Gravity Probe B)

In high-density environments, the vacuum is "loaded" by matter-energy ρ_m , which saturates its tensor susceptibility. This state of mechanical satiation manifests as the "viscosity" of the medium, traditionally interpreted as frame-dragging. We correlate the observed Lense-Thirring precession Ω_{LT} to the vacuum's mechanical propagation floor v_γ . The linear drift velocity at Earth's radius $R_\oplus$ is:

$$v_{drift} = \Omega_{LT} \cdot R_\oplus \approx 4.2 \times 10^{-8} \text{ m/s} \quad (7)$$

Consistent with the derivation in Section 3.2, the propagation floor v_γ is determined by the local refractive index:

$$v_\gamma = c \sqrt{\frac{8\pi G}{8\pi G + c^2}} \approx 4.09 \times 10^{-5} \text{ m/s} \quad (8)$$

The alignment of these magnitudes (both significantly sub-relativistic and determined by the G/c^2 ratio) suggests that frame-dragging is the physical displacement of a satiated, high-impedance vacuum medium. The 8.6% Hubble Tension further implies that as G recovers in low-density environments, this mechanical coupling increases, offering a unified explanation for galactic rotation anomalies.

5.6 5.2 Alignment of Observed vs. Theoretical Drift

To validate the mechanical model, we compare the theoretical propagation floor v_γ to the empirical drift v_{drift} measured by Gravity Probe B.

$$v_\gamma = c\sqrt{\frac{8\pi G}{8\pi G + c^2}} \approx 4.09 \times 10^{-5} \text{ m/s} \quad (9)$$

$$v_{drift} = \Omega_{LT} \cdot R_\oplus \approx 4.2 \times 10^{-8} \text{ m/s} \quad (10)$$

We define the **Vacuum Coupling Efficiency** (κ) as:

$$\kappa = \frac{v_{drift}}{v_\gamma} \approx 10^{-3} \quad (11)$$

This indicates that in the Earth's high-density satiated environment, the vacuum dragging efficiency is 0.1%. The observed 8.6% Hubble Tension provides the recovery gradient: as density ρ_m decreases, κ increases, providing the additional centripetal coupling observed in galactic rotation curves without requiring dark matter particles.

5.7 The Universal Floor: Unruh Satiation

The Unruh effect predicts a thermal bath $T = \hbar a / (2\pi c k_B)$ for an accelerating observer. We propose this temperature represents the stochastic noise floor of the vacuum at the limit of mechanical satiation. At the Milgromian acceleration constant $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$, we equate the energy of the vacuum satiation velocity to the Unruh thermal energy. This indicates that the suppression of G is maintained as long as the gravitational potential energy of the vacuum is in equilibrium with the Unruh radiation pressure. This provides a physical basis for the "Newtonian floor," preventing the gravitational coupling from diverging in high-acceleration regimes.

Mathematical Comparison: At the Milgromian acceleration constant $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$, we equate the energy of the vacuum satiation velocity to the Unruh thermal energy:

$$\frac{1}{2}mv_\gamma^2 \sim k_B T_{Unruh} \quad (12)$$

This indicates that the "clamping" of G occurs exactly when the gravitational potential energy of the vacuum reaches equilibrium with Unruh radiation pressure, preventing G from diverging.

5.8 The Galactic Solution: Recovery of Ground-State Tension

Dark Matter is reinterpreted as the vacuum's return to its **Unsaturated Ground State**. As ρ_m drops in the galactic halo, G scales upward as the impedance drops.

Mathematical Comparison: To maintain a flat rotation curve, the required centripetal acceleration is $a = v^2/r$. Standard theory adds Dark Matter (M_{DM}); our model scales G :

$$\frac{v^2}{r} = \frac{G(\rho)M_{baryonic}}{r^2} \quad (13)$$

As ρ_m drops below the critical threshold ρ_{crit} , $G(\rho)$ increases by a factor of ≈ 10 , compensating for the “missing” mass. The “Dark Matter” effect is the vacuum recovering its restorative stiffness (c^2) in the absence of mechanical load.

Environment	Density	Effective G	Effect
Solar System	High	G -Newton	Standard Orbits
Galactic Halo	Low	$G \rightarrow G$ -max	Flat Rotation Curves
Cosmic Void	$\rightarrow 0$	G -max	Enhanced Lensing

6 Discussions

This mechanical model predicts that lensing in low-density voids is enhanced, aligning with discrepancies in lensing signals from certain cosmic voids noted by Sanchez et al. (2017)⁶, and further explains the following observations:

6.1 The Shapiro Delay (The “Mechanical” Proof)

This is perhaps the most direct observation of a “propagation floor” in action. Irwin Shapiro first predicted and measured that radio signals passing near the Sun take longer to return than expected. **The Observation:** Radar pulses sent to Venus and Mercury “slow down” as they pass through the high-density gravitational well of the Sun. **Interpretation of this Model:** As the signal enters a high-density region (ρ_m increases), the vacuum reaches its satiation limit. The impedance rises, and the propagation speed drops toward the local “floor.” The “delay” is the physical manifestation of the wave traveling at a reduced velocity through a satiated medium.

6.2 Gravitational Redshift (Pound-Rebka Experiment)

If the propagation speed of the field is linked to its tension (G), then moving through different densities should change the wave’s energy. **The Observation:** Light escaping a gravitational well (moving from high ρ_m to lower ρ_m) shifts toward the red end of the spectrum. **Interpretation of this Model:** This is a transition from a satiated, stiff vacuum to an unsatiated, pliant vacuum. The change in G and propagation velocity causes the wave to “stretch,” as it is moving from a high-impedance “floor” back toward its ground-state maximum speed.

6.3 Cosmic Voids and Lensing Anomalies

Observations of cosmic voids (large regions of very low density) have shown discrepancies in gravitational lensing—the way light bends around matter. **The Observation:** Some studies (like Sanchez et al., 2017) suggest that lensing signals in voids are stronger than predicted by standard GR. **Interpretation of this Model:** In these low-density voids,

the vacuum is unsatiated. G is higher than the Newtonian floor, and the propagation speed is closer to its true ground-state maximum. This increased "tensor susceptibility" causes light to deviate more than it would in a "stiff" Solar System environment.

6.4 The Variable Speed of Light (VSL) Theories

In cosmology, theorists like João Magueijo have proposed that c was significantly higher in the early universe to solve the "Horizon Problem" (why the universe looks uniform).

Observation/Theory: This addresses the lack of thermal equilibrium in the CMB.

Interpretation in your Model In the extremely early universe, before the "satiation" caused by large-scale structure formation, the vacuum impedance was lower, allowing for a much higher propagation speed. This aligns with the idea that G and c are tied to the mechanical state of the field.

Author Information

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